

**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	7 July 2026
Team ID	SWTID-2026-5632
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Guidelines:

1. Include all the processes (As an application Logic/ Technology Block)
2. Provide infrastructural demarcation (Local)
3. Indicate external interfaces (if any)
4. Indicate data storage and components/services
5. Indicate Interface to machine learning models (if applicable)



TECHNICAL ARCHITECTURE

The deliverable shall include the architectural diagram as below and the information as per the table-1 & table-2.

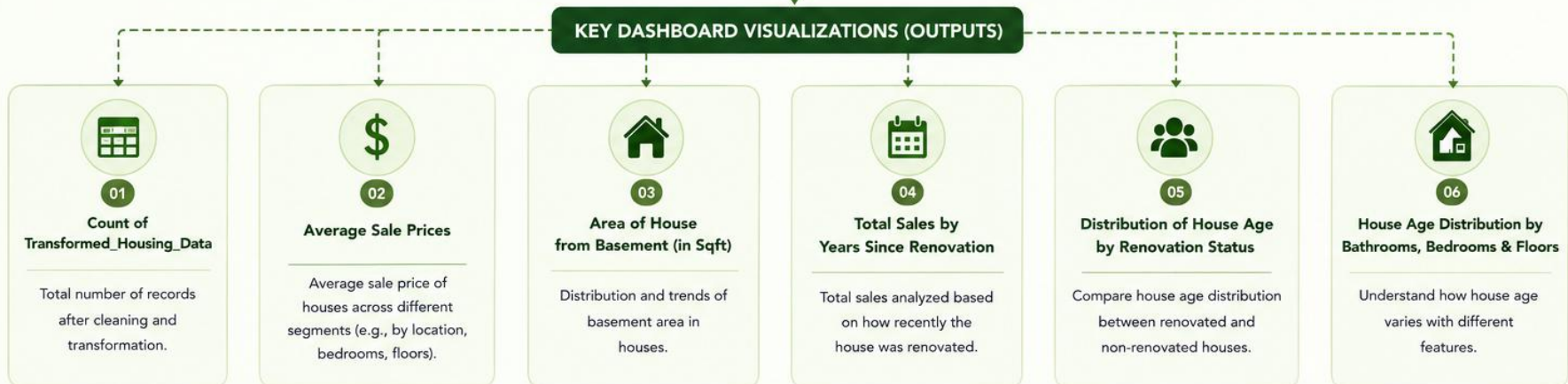
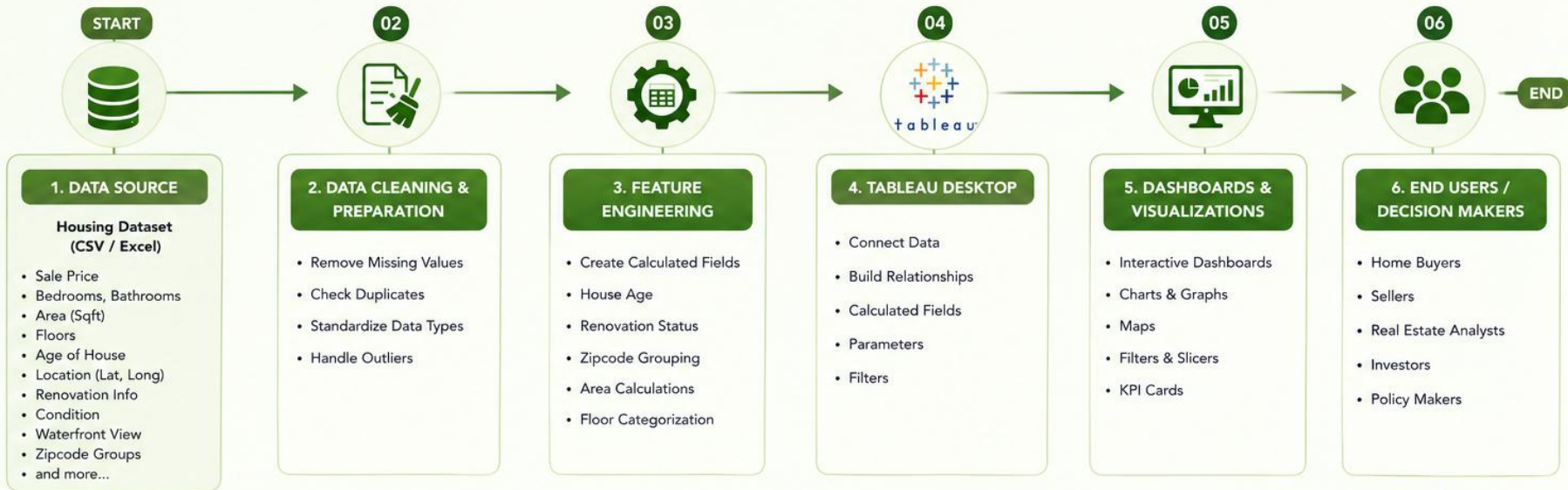


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard used to explore housing market trends.	Tableau Dashboard
2.	Application Logic-1	Data cleaning and preprocessing before analysis.	Tableau Filters
3.	Application Logic-2	Feature engineering and calculated fields.	Tableau Calculated Fields
4.	Application Logic-3	Interactive analysis using filters, parameters and charts.	Tableau Desktop
5.	Database	Stores the housing dataset for analysis.	CSV / Excel Dataset
6.	Cloud Database	Not used in this project.	N/A
7.	File Storage	Stores dataset and Tableau workbook.	Local Storage (.csv, .xlsx, .twb/.twbx)
8.	External API-1	Not used.	Not Applicable
9.	External API-2	Not used.	Not Applicable
10.	Machine Learning Model	Machine learning is not implemented in this project.	Not Applicable
11.	Infrastructure (Server / Cloud)	Project developed and executed on a local computer.	Windows PC, Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Dataset prepared using Microsoft Excel	Microsoft Excel

2.	Security Implementations	Read-only project files with local system security	Windows File Permissions
S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Architecture can easily support additional housing records, dashboard sheets and calculated fields without redesigning the system.	Tableau Public, CSV/Excel Dataset
4.	Availability	The dashboard is available whenever the Tableau workbook and dataset are opened on the local computer. Internet is only required for publishing to Tableau Public (if used).	Tableau Public
5.	Performance	Efficient processing of housing data using Tableau's in-memory engine, filters, calculated fields and optimized visualizations for faster dashboard response.	Tableau

References:

<https://c4model.com/> <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/> <https://www.ibm.com/cloud/architecture> <https://aws.amazon.com/architecture>
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>